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GCE-UQ: QUANTIFYING AND DECOMPOSING UNCERTAINTY IN GRAPH COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Graph counterfactual explanations (GCEs) are increasingly used to interpret Graph Neural Networks, yet existing methods ignore the *embedding-equivalent graph set* that models implicitly treat as task-indistinguishable. We formalize the resulting *Graph Counterfactual Explanation Uncertainty* (GCEU) and derive its exact decomposition into *Topological* and *Edit* uncertainties. We introduce **GCE-UQ**, a framework that *quantifies*, *decomposes*, and *mitigates* uncertainty for GCEs. The three uncertainty terms are differentiable and can thus be plugged into existing objectives. Experiments on synthetic and real-world datasets show that our measures reliably distinguish genuine counterfactual edits from adversarial perturbations. Furthermore, incorporating our uncertainty regularizer into existing GCE frameworks notably improves the robustness of the generated counterfactuals.

Index Terms— Graph Neural Networks, Counterfactual Explanations, Uncertainty Quantification, Robustness

1. INTRODUCTION

Graph Neural Networks (GNNs) have become essential for learning from relational data [1]. In critical applications like medical diagnosis [2, 3, 4], intrusion detection [5, 6], and fraud detection [7, 8], users need to understand why a GNN makes specific predictions. Wrong decisions in these domains can have serious consequences.

Graph counterfactual explanations (GCEs) provide an intuitive approach to understanding GNN decisions. They identify minimal graph changes that flip the model's prediction [9, 10]. For example, in loan approval, a user's creditworthiness can be assessed through their transaction network, represented as a graph where nodes are accounts and edges represent money transfers. A GCE might reveal that if just two specific transfer connections had not existed, the model would approve rather than reject the loan. This analysis clarifies what changes are necessary to achieve a different decision.

Existing GCE methods [11, 12, 13, 10] focus on the observed graph G_{obs} and produce an edit ΔG that changes its predicted label. Yet a GNN embeds many task-equivalent graphs—those differing only by minor, task-irrelevant topological changes—into a tight region of the embedding space, which we denote as the *embedding-equivalent graph set* S_{emb}. Because the model treats every graph in S_{emb} similarly, a reliable counterfactual should also succeed on most graphs in S_{emb}, not just on G_{obs}. Ignoring this can lead to explanations that conflict with the model's own generalization behavior.

Figure 1 illustrates the risk in a loan-approval setting. Each point is a user's transaction subgraph, and the curve is the GNN's decision boundary. In practice, unknown transactions can make the true graph

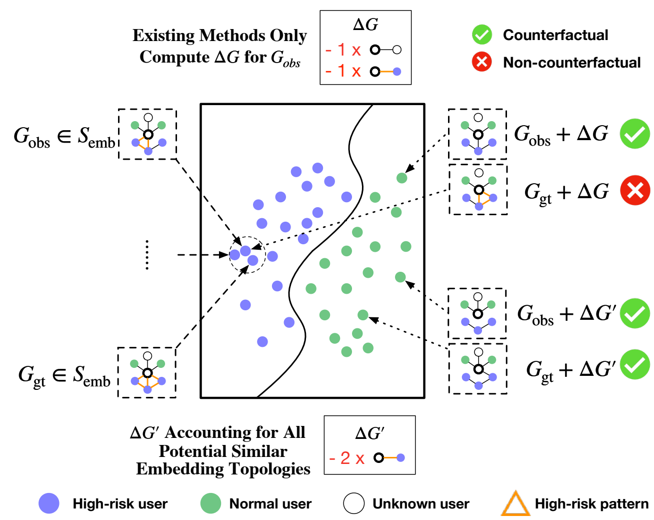

Fig. 1: Illustration of graph counterfactual uncertainty in a loan-approval task. G_{obs} is the observed transaction subgraph, while G_{gt}, the "true graph", represents the applicant's full transaction history (including as-yet unobserved interactions) that the model also classifies correctly. The edit ΔG exhibits high uncertainty: it flips G_{obs} but fails on G_{gt}. By considering the full embedding-equivalent graph set S_{emb}, $\Delta G'$ has low uncertainty and succeeds on both graphs.

G_{gt}, which the GNN still classifies correctly. The edit ΔG designed for G_{obs} fails on G_{gt}. A loan officer who relies on the explanation from ΔG may thus grant credit to a high-risk applicant by mistake. $\Delta G'$, by taking into account all graphs in S_{emb}, flips both G_{obs} and G_{gt}, yielding an explanation that stays valid.

We term this phenomenon graph counterfactual uncertainty (GCEU), referring to the variability in model predictions when applying the same counterfactual edit to G_{obs} versus other equivalent graphs in S_{emb}. In Figure 1, ΔG exhibits high uncertainty by failing on G_{gt}, whereas $\Delta G'$ shows low uncertainty by successfully flipping both graphs. In sensitive applications, handling this uncertainty is critical to maintain user trust and avoid harmful errors.

Challenges. Despite the need to quantify and reduce GCEU, several obstacles stand in the way. First, there is no formal definition of GCEU that captures the variability of prediction across the infinite and unknown set S_{emb}, making measurement and optimization challenging. Second, classical uncertainty quantification focuses on model confidence for fixed inputs. These methods do not capture how a counterfactual edit's effect varies across different members of S_{emb}, so they cannot be applied directly. Third, GCEU arises from

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diverse sources: model-indistinguishable, task-irrelevant topological variations within $S_{\rm emb}$, and the risk that an edit's observed effect on $G_{\rm obs}$ does not reflect its typical effect over $S_{\rm emb}$. Decomposing GCEU into these sources is necessary to target the dominant cause.

GCE-UQ Framework. To address these challenges, we propose our Graph Counterfactual Explanation Uncertainty Quantification (GCE-UQ) Framework with four key components. We begin by introducing a formal definition of GCEU. Next, the GCE-UQ Framework employs an efficient Metropolis–Hastings MCMC sampler that approximates draws from $S_{\rm emb}$ under embedding, structural, and label constraints, enabling accurate estimation of GCEU. Third, we derive an information-theoretic decomposition that splits total graph counterfactual uncertainty (GCEU) into two parts: topological uncertainty (TopoU), which quantifies variations from model-indistinguishable, task-irrelevant topological variations in $S_{\rm emb}$, and edit uncertainty (EditU), which captures the inconsistency between an edit's observed effect on $G_{\rm obs}$ and its typical effect over $S_{\rm emb}$. Finally, since all three uncertainty measures (GCEU, TopoU, and EditU) are differentiable, they can be used as plug-in regularizers for most existing counterfactual explanation methods, guiding edits toward lower uncertainty.

Contributions. Our contributions are threefold: (i) we give the first formal definition of Graph Counterfactual Explanation Uncertainty (GCEU) and an exact decomposition; (ii) we introduce GCE-UQ, which quantifies and decomposes GCEU and offers a differentiable uncertainty regularizer that integrates with existing GCE methods to guide low-uncertainty counterfactuals; and (iii) on synthetic and real-world benchmarks, we show that these metrics separate true counterfactual edits from adversarial perturbations and that the regularizers improve GCE robustness.

2. GCE–UQ FRAMEWORK

This section introduces our **G**raph **C**ounterfactual **E**xplanation **U**ncertainty **Q**uantification (GCE–UQ) framework. We follow Sec. 1: first formalising the embedding–equivalent graph set, then defining GCEU and its decomposition, and finally discussing differentiability, compatibility, complexity, and limitations.

2.1. Problem Setup and Counterfactual Objective

2.1.1. Tasks and model.

An attributed graph is $G = (V, E, \mathbf{X})$ with node features $\mathbf{X} \in \mathbb{R}^{|V|\times d}$. We use an L-layer GNN f_θ for both graph and node classification. For graph classification, the model outputs class probabilities $f_\theta(G)$ and prediction $\hat{y}(G) = \arg\max_c f_\theta(G)_c$. For node classification, we form an L-hop subgraph G_{v^*} around a target node v^* and predict $\hat{y}(v^*)$ from $f_\theta(G_{v^*})$. We denote by $G_{\rm obs}$ the observed input (graph or subgraph) whose prediction we explain.

2.1.2. Counterfactual objective.

A counterfactual is a small edit ΔG applied to $G_{\rm obs}$ that changes the predicted label. We restrict ΔG to the topology: adding or removing edges; node features remain fixed. Let $\mathrm{cost}(\Delta G)$ be the number of edge flips (or a weighted variant). We seek a valid edit with minimum cost, written as $\min_{\Delta G}\ \mathrm{cost}(\Delta G)$ s.t. $\hat{y}(G_{\rm obs} + \Delta G) \neq \hat{y}(G_{\rm obs})$.

2.2. Embedding–Equivalent Graph Set $S_{\rm emb}$

Intuitively, $S_{\rm emb}$ collects all latent graphs that the trained GNN perceives as *task–equivalent* to the observed graph $G_{\rm obs}$. We formalise this intuition via three constraints that leave the model's decision unchanged while allowing benign topological variations.

Definition 1 (Embedding–Equivalent Graph Set)**.** *Let $G_{\rm obs} = (V, E_{\rm obs}, \mathbf{X})$ be the observed graph. Denote by $\mathbf{h}(G)$ the embedding produced by the trained GNN and by $\hat{y}(G)$ its predicted label. For tolerances $\delta_{\rm emb}, \delta_A > 0$, define the embedding-equivalent graph set*

$$S_{\rm emb} = \left\{ g = (V, E, \mathbf{X}) \;\middle|\; \substack{\|\mathbf{h}(g)-\mathbf{h}(G_{\rm obs})\|\le\delta_{\rm emb} \\ \hat{y}(g)=\hat{y}(G_{\rm obs}) \\ d_A(E,E_{\rm obs})\le\delta_A} \right\}.$$

In Def. 1, the first constraint bounds representation drift in the embedding space; the second preserves the model's semantic decision (label consistency); and the third caps the edge-edit distance from the observed topology. Together, these conditions ensure that every $g \in S_{\rm emb}$ is behaviourally indistinguishable from $G_{\rm obs}$ in both prediction and latent geometry, while still allowing a rich space of latent graphs.

2.3. Graph Counterfactual Uncertainty

Given a candidate counterfactual edit ΔG produced by any GCE method, we quantify how uncertain its effect is across $S_{\rm emb}$.

Definition 2 (Graph Counterfactual Uncertainty (GCEU))**.** *Let $P(g)$ be the uniform counting* probability *measure on $S_{\rm emb}$. For a counterfactual edit ΔG, its* graph counterfactual uncertainty *is*

$$\mathrm{GCEU}(\Delta G) \;=\; \mathbb{E}_{g\sim P(g)}\Big[\mathrm{KL}\big(P(Y^{\Delta G} \mid g) \;\|\; P(Y^{\Delta G} \mid G_{\rm obs})\big)\Big],$$

where $Y^{\Delta G}$ denotes the model prediction after applying ΔG.

GCEU is **low** when, for *most* graphs $g \in S_{\rm emb}$, the predictive distribution after editing is close (in KL divergence) to that on the observed graph $G_{\rm obs}$; it becomes **high** when the behaviour on $G_{\rm obs}$ is atypical relative to the bulk of $S_{\rm emb}$.

We estimate the expectation in GCEU by sampling feasible graphs from $S_{\rm emb}$ with a Metropolis–Hastings random walk that flips a single edge; proposals that violate $S_{\rm emb}$ are rejected. The symmetric proposal and the connectivity induced by single-edge flips make the chain ergodic with the uniform stationary measure on $S_{\rm emb}$ [14]. After burn-in, we apply thinning (keep every k-th state to reduce autocorrelation) and then average the KL term over the retained samples, which yields the estimate of GCEU.

2.4. Information–Theoretic Decomposition

Theorem 1 (GCEU Decomposition)**.** *For any counterfactual edit ΔG,*

$$\mathrm{GCEU}(\Delta G) = \underbrace{I\big(Y^{\Delta G}; g\big)}_{\textit{TopoU}} + \underbrace{\mathrm{KL}\big(P(Y^{\Delta G}) \,\|\, P(Y^{\Delta G} \mid G_{\rm obs})\big)}_{\textit{EditU}}, \tag{1}$$

where $I(\cdot;\cdot)$ denotes mutual information and

$$P(Y^{\Delta G}) = \mathbb{E}_{g\sim P(g)}[P(Y^{\Delta G} \mid g)]. \tag{2}$$

Sketch. Start from the GCEU expression and apply the identity for the KL divergence, $\mathrm{KL}(P\|Q) = \mathbb{E}_P\Big[\log\frac{P}{Q}\Big]$, to its inner term. Insert the marginal $P(Y^{\Delta G})$ and split the logarithm:

$$\log\frac{P(Y^{\Delta G} \mid g)}{P(Y^{\Delta G} \mid G_{\rm obs})} = \log\frac{P(Y^{\Delta G} \mid g)}{P(Y^{\Delta G})} + \log\frac{P(Y^{\Delta G})}{P(Y^{\Delta G} \mid G_{\rm obs})}.$$

After taking the expectation over g, the expression separates into two parts. The first is the standard mutual information form and equals $I(Y^{\Delta G}; g)$. The second no longer depends on g and equals $\mathrm{KL}\big(P(Y^{\Delta G}) \,\|\, P(Y^{\Delta G} \mid G_{\text{obs}})\big)$. Combining them gives (1). $\square$

TopoU captures the intrinsic variance due to task–irrelevant topological choices within S_{emb}; **EditU** quantifies how atypical the effect observed on G_{obs} is relative to the average over S_{emb} and can thus be minimised by choosing more representative edits.

3. EXPERIMENTS

3.1. Experimental Setup

Datasets. We evaluate on five benchmarks across two tasks. *Node classification:* We use synthetic datasets BA-Shapes, Tree-Cycles, and Tree-Grid [15], which embed discriminative motifs into base structures with labels assigned by structural roles. A three-layer GCN achieves test accuracies of 98.6%, 95.5%, and 91.1%, respectively. *Graph classification:* We evaluate on real-world datasets Mutagenicity [16], NCI1 [17], where nodes represent atoms and graphs represent molecules with labels determined by chemical properties. Using three-layer GCN with mean pooling, we achieve test accuracies of 80.7% and 76.2%, respectively.

Counterfactual Methods. In the main text, we primarily focus on CF-GNNExplainer [11] as the base counterfactual generator. Note that our GCE-UQ framework is general and can be applied to any counterfactual explanation method that is compatible with the assumptions in Definitions 1-2. Additional experimental results on more datasets, alternative counterfactual explanation methods, and detailed implementation specifics are provided in the supplementary material (`https://github.com/chenghuaguo/GCE-UQ/`).

3.2. Evaluating Uncertainty Discriminative Power

We evaluate whether our uncertainty metrics can distinguish *genuine counterfactual edits* from *adversarial perturbations*. This comparison validates our uncertainty measures based on fundamental differences between these two types of edits. Adversarial perturbations must cause misclassification and exploit non-robust features that are predictive for the model but lack human-interpretable patterns [18, 19]. Given our models' high test accuracies, adversarial attacks necessarily target model vulnerabilities rather than robust decision boundaries. In contrast, counterfactual explanations seek minimal edits without requiring misclassification [18]. The reliance of adversarial perturbations on model-specific vulnerabilities should make them more sensitive to structural variations in S_{emb}, resulting in higher uncertainty compared to counterfactual explanations.

For each test sample, we generate a *counterfactual edit* ΔG_{CF} and construct an *adversarial topological edit* ΔG_{PGD} using the PGD attack of [20] with the same edit budget. On synthetic node datasets, we restrict the attack to edges outside the class motif (PGD-Motif) to ensure that adversarial perturbations cause misclassification without altering the ground-truth node labels, which aligns with the definition of adversarial attacks [18]. We also consider a variant that preserves all counterfactual edges (PGD-CF), which is the only option for real-world datasets since no canonical motif exists [21]. We execute attacks only when a valid counterfactual is first obtained, then compute GCEU, EditU, and TopoU for both ΔG_{CF} and ΔG_{PGD}.

Figure 2 reports uncertainty score distributions using violin plots for the synthetic datasets, where the attacker is restricted to *non-motif* edges (PGD-Motif). Counterfactual edits concentrate at relatively lower uncertainty values compared to PGD edits for all three measures, showing narrower distributions. In contrast, PGD edits exhibit broader distributions that extend to higher uncertainty values.

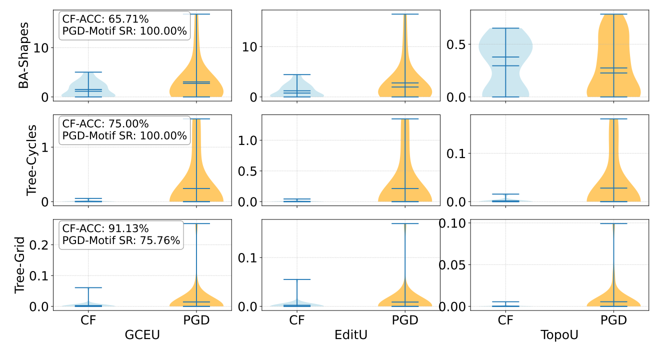

Fig. 2: Uncertainty score distributions on synthetic node-classification datasets using violin plots. Each row represents a dataset, and each column shows a different uncertainty metric. Counterfactual edits (CF) are shown in blue; PGD attacks confined to *non-motif* edges are shown in orange. CF-ACC indicates the fraction of test nodes for which a counterfactual is found; PGD-Motif SR shows the attack success rate when such a counterfactual exists.

Figure 3 repeats the experiment with a different constraint: the attacker may flip any edge *except* those selected by the counterfactual (PGD-CF). Here, overall GCEU distributions between CF and PGD edits appear more similar than in Figure 2, highlighting the importance of uncertainty decomposition. Examining individual components, TopoU shows minimal differences between CF and adversarial edits across all datasets, and notably, TopoU values are orders of magnitude smaller than EditU values. This indicates uncertainty arising from PGD-CF attacks is primarily driven by EditU rather than intrinsic topological variance, suggesting that these adversarial perturbations achieve their effect through atypical structural modifications rather than exploiting inherent graph ambiguities.

3.3. Evaluating Uncertainty Regularization for GCE Robustness

We investigate whether uncertainty regularization can improve counterfactual explanation quality when the observed graph G_{obs} differs from the unknown true graph G_{gt} through small prediction-preserving perturbations. This scenario reflects the realistic setting where the input graph contains topological noise that does not affect the model's primary prediction but may destabilize explanation generation. By incorporating our uncertainty metrics (GCEU, EditU, TopoU) as regularization terms in the optimization objective of perturbation-based [10] GCE methods, we test whether prioritizing low-uncertainty explanations enhances explanation robustness.

Our noise injection protocol systematically varies the noise level $\varepsilon \in \{1, \ldots, 20\}\%$ by randomly flipping that fraction of edges to construct $G_{\text{gt}}^{(\varepsilon)}$. When the predicted class changes, we resample until $\hat{y}(G_{\text{gt}}^{(\varepsilon)}) = \hat{y}(G_{\text{obs}})$, ensuring that noise affects only structural representation rather than semantic content. We then apply the original counterfactual edit to both the clean and noisy graphs, yielding $G_{\text{obs}} + \Delta G$ and $G_{\text{gt}}^{(\varepsilon)} + \Delta G$. To account for randomness in edge selection, we repeat this process across ten random noise instantiations and average the resulting measurements.

We primarily evaluate explanation robustness through CF-ACC, which measures the fraction of instances where the counterfactual maintains the same predicted class under noise: $\frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[\hat{y}_i(G_{\text{obs}}+$

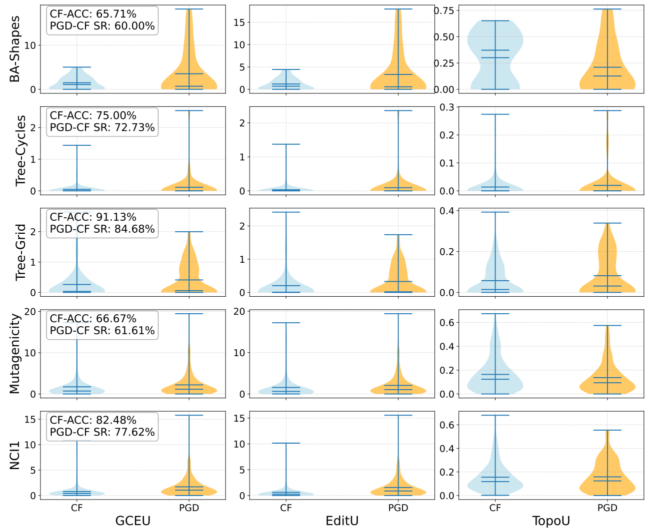

Fig. 3: Uncertainty score distributions when the attacker is prohibited from altering edges chosen by the counterfactual (PGD–CF). The layout matches Figure 2. For the molecular graph datasets this is the only attack variant because no canonical motif is defined.

$\Delta G) = \hat{y}_i(G_{\text{gt}}^{(\varepsilon)} + \Delta G)]$. This metric directly reflects whether uncertainty regularization produces more stable GCEs.

Table 1 demonstrates that uncertainty regularization significantly improves counterfactual robustness across datasets and noise levels. All three uncertainty metrics achieve statistically significant gains over vanilla baselines in the majority of configurations, with improvements becoming more pronounced at higher noise levels–precisely when robust explanations are most critical.

Crucially, **no single uncertainty metric dominates universally**, validating the necessity of our decomposed approach. The performance patterns reveal clear dataset-specific preferences: GCEU achieves the largest gains on BA-Shapes at low noise levels (90.9% vs. 72.8% at 1% noise), EditU consistently outperforms others on Mutagenicity across all noise levels, and TopoU dominates Tree-Cycles especially at higher noise levels. These distinct advantages demonstrate that **decomposed uncertainty regularization** enables domain-adaptive selection, allowing practitioners to choose the most effective component for their specific graph types rather than settling for a one-size-fits-all approach. Additional metrics and complete robustness curves are provided in the supplementary material.

4. RELATED WORK

UQ for Counterfactual Explanations. Delaney *et al.* [22] evaluate standard UQ tools on samples already edited by counterfactual methods, revealing pitfalls in detecting out-of-distribution points, but do not quantify uncertainty of the *edit itself*. Recent advances include PROBE [23], which incorporates failure probabilities into recourse optimization for tabular data, and QUCE [24], which quantifies uncertainty along gradient paths for generative counterfactuals. However, these approaches either target non-graph domains or provide monolithic uncertainty scores without decomposing uncertainty sources. GCE-UQ is the first to decompose uncertainty directly on counterfactual actions in graph domains.

Robustness of GCE. Existing robustness studies focus on *consistency of explanations* under input perturbations [25] or model changes [26]. Chen *et al.* [27] further explore explanation consistency under noisy

Table 1: CF-ACC comparison across noise levels. **Bold** indicates best. †/* denote significant/highly significant improvement over *Vanilla* by a one-sided Welch t-test (p<0.05/0.01).

Dataset	Noise Level	CF-ACC ↑ Vanilla	GCEU	EditU	TopoU
BA-Shapes	1%	72.8	**90.9***	80.3*	85.3*
	5%	43.0	**57.7***	54.3*	52.1*
	10%	30.0	37.5†	38.5*	**39.4***
	15%	25.2	**34.5***	25.7	30.4
	20%	15.6	**24.5***	16.7	21.9*
Tree-Cycles	1%	97.0	**98.2**†	97.7	97.6
	5%	73.5	77.2†	**77.3**†	76.1
	10%	54.7	56.6	56.2	**61.3***
	15%	40.6	45.7†	44.8†	**46.9***
	20%	35.2	39.3†	38.6†	**43.1***
Tree-Grid	1%	86.8	88.4	89.5*	**90.0***
	5%	50.4	**58.1***	56.9*	57.2*
	10%	35.4	42.6*	40.4*	**42.6***
	15%	24.4	**33.9***	33.6*	31.6*
	20%	18.7	**26.7***	26.2*	24.4*
Mutagenicity	1%	57.8	57.3	**59.3**	59.0
	5%	27.5	29.7†	**32.7***	27.4
	10%	17.2	20.8*	**22.2***	18.5
	15%	12.5	17.0*	**18.3***	15.6*
	20%	10.7	14.0*	**14.8***	12.9*
NCI1	1%	56.2	56.4	56.5	**62.9***
	5%	29.3	31.2†	34.1*	**34.7***
	10%	20.8	23.4*	**24.4***	23.8*
	15%	15.1	18.0*	19.6*	**19.8***
	20%	11.8	15.1*	**16.9***	15.2*

inputs. In contrast, GCE-UQ regularizes GCE algorithms to return *one* edit whose predictive effect remains stable across task-irrelevant variations, providing uncertainty-aware robustness rather than post-hoc consistency evaluation.

5. CONCLUSION

We introduced GCE-UQ, the first framework to formally define, quantify, and mitigate uncertainty in GCEs. Our key contribution is the formalization of GCEU and its exact decomposition into TopoU and EditU, which captures distinct sources of explanation uncertainty. The GCE-UQ framework employs an efficient MCMC sampler to approximate the embedding-equivalent graph set and provides differentiable uncertainty measures that integrate seamlessly with existing GCE methods as regularization terms. Experimental validation demonstrates that our uncertainty metrics reliably distinguish genuine counterfactual edits from adversarial perturbations. Uncertainty regularization significantly improves explanation robustness under structural noise, with gains becoming more pronounced at higher noise levels. The decomposed uncertainty components show dataset-specific advantages, enabling practitioners to select the most effective regularizer for their domains. Future work will extend UQ to hybrid GCEs involving both topological and feature modifications, and explore GCE uncertainty quantification for dynamic graphs.

6. ACKNOWLEDGMENTS

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